

In the classical limit of the RTLI

$$\mathcal{H} = - \sum_{i=1}^L |J_i| \sigma_i \sigma_{i+1} - \sum_{i=1}^L h_i \sigma_i$$

with L even, J_i and h_i random variables. The parameters J_i are uniform in $[0, 1]$ and h_i are uniform in $[-\frac{h}{2}, \frac{h}{2}]$. This is just a regular RFIM in 1d.

In the classical limit of the RATIC

This is antiferromagnetic in a constant magnetic field. Using a Mattis transformation it becomes a ferromagnetic model in an alternate field

$$\mathcal{H} = - \sum_{i=1}^L |J_i| \sigma_i \sigma_{i+1} - \sum_{i=1}^L (-1)^i h \sigma_i$$

This is not regular RFIM, since the fields alternate.

Conclusion

It is different and the trick of the RATIC cannot be used as is in the RTLI ...

Do you agree?